The Heist: Zero Hour Challenge Book (Final)

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At a glance

This is the final challenge set for 30 to 40 first-year engineering students: 42 challenges that need logic, not syntax, and every one of them runs inside the game screen. It replaces the earlier 42-challenge list and the SQL-based mix in the build specification. Everything about money, slots and schedule is rescaled for a maximum of 40 players.

Item	Value
Players	30 to 40 (everything is sized for 40)
Event length	180 minutes, schedule unchanged from the build specification
Total money	₹4,000 and no feature creates money
Cap for one player	₹800 (Home plus Bank). At the cap the player stays in the game: new pay is trimmed, but the money can still be stolen
Raid crew	Exactly 10 players, 5 must pass the 2-answer gate
Challenges	42: 8 Code, 7 Debug, 14 CTF, 13 Puzzle
Languages shown	C 5, Python 5, Java 4, plus one race challenge served in all three. No C++ and no SQL
Checking	In the browser. The server only checks the claim (slot, reservation, limits) and pays

Rules for first-year level

- **Logic first.** A player needs loops, conditions, lists and careful reading. No data structures beyond lists and arrays, no recursion, no SQL.
- **Expert means more steps, not a harder topic.** Trees and graphs only appear as drawings to read.

- **CTF is logic.** Ciphers with every table on screen, clue deduction, log reading and number machines. No code reading in CTF.
- **Nothing leaves the screen.** Lookup tables, a scratchpad and a calculator live inside every challenge.
- **Variants.** The client builds each player's numbers, words and clues from a seed (player ID plus challenge ID), so two players never see the same data and a player sees the same data again.
- **Guess lockout.** Three wrong answers lock that challenge for 60 seconds. Typed answers are otherwise easy to brute force.

The money loop

Money is only worth something while it is safe, so every rule pushes players to stay in play and to risk each other.

- **Earn.** Pay lands at the player's Home. Whenever a player is away solving a challenge, that Home is exposed.
- **Steal.** A thief solves a lock-pick puzzle at the Home door. Easy (BG-01) takes 5% of the Home money, hard (BG-02) takes 10%, difficult (BG-03) takes 20%, with maximums of ₹10, ₹25 and ₹50.
- **Bank it.** The Central Bank is safe from burglary but charges a holding fee every minute the money sits there. The fee is a percentage of the balance (0.1% per minute to start) and feeds the next raid prize.
- **Raid.** A crew of 10 can raid the Bank. It is the one team challenge, and the hardest.
- **No exit at the cap.** A player who reaches ₹800 keeps playing. They must defend, bank or steal, because their money is still at risk.
- **No early exit.** Checkout opens only at minute 150, so leaving early is not a way to lock in a score. This is a proposed rule, listed under open decisions.

How challenges run in the app

Every challenge opens in one screen over the map, and the player never needs anything outside it. Every screen has a scratchpad (free text) and a mini calculator. On top of that, each type has its own workbench.

Type	Challenges	Workbench in the app
Code reading (C, Python, Java)	ST-01, SD-01, SD-02, SD-03, RC-01, GH-03, BR-02, BL-01	Code panel with line numbers and tap-to-highlight, plus a trace table (a row per loop pass, a column per variable). The player types what the program prints.
Debug	ST-02, SD-04, SD-05, EX-03, RC-02, BR-08, BL-05	Tap the buggy lines, pick the fix from the options, then press "Run fixed code" to see the output change. The output comes from the seed, so no compiler is needed.
Cipher CTF	ST-03, SD-08, RC-03, GH-01, BR-03, BL-02	A cipher bench with the ciphertext, a Caesar shift wheel, an Atbash tool, a reverse tool and a letter-by-letter substitution board, all with a live preview. The logic is deciding which steps to use and in which order.
Clue deduction	SD-06, BR-06, BR-07	An elimination board: the word list or digit slots, tap to strike out, a clue list with ticks, and a notes box.
Log reading	SD-07, BR-04	A log viewer with a search box, filters by user, IP and status, and tap-to-pin rows into a side panel.
Number machine	EX-04, BL-04	A diagram of the step chain with 5 test runs on any input. The player works backwards using the runs and the calculator.
Fake portal	EX-05	A mock notice board and login form drawn inside the challenge screen.

Type	Challenges	Workbench in the app
Grid and diagram puzzles	EX-01, EX-02, BR-01, BR-05, BR-09, BL-03	The player moves a token, taps nodes to mark a chain or a loop, or drags tasks into order. The page checks the result.
Existing puzzles	SD-09, SD-10, BG-01, BG-02, BG-03, GH-02, BR-10	Already interactive: lights-out board, statement cards, keypad, maze, fuse box, higher/lower dialogue, constraint board.

Skinning

Reuse the Raid Circle modal style, including the "VEYRA SYSTEMS / VERIFIED UPLINK" header. Change only the header label per type (`CONTRACT / CODE` , `CONTRACT / CTF`) and the accent colour. Each contract card shows the client, a two-line briefing, a one-sentence objective, the reward and time, and a stamp with difficulty pips.

Where challenges live

The 18 paid jobs keep the station mapping of the build specification, so the boards, waves and Expert schedule need no change. Only the SQL stations change: the Data Centre now hosts reading puzzles, and the SQL Terminal has no SQL left, so relabel it "Code Review Desk" (label change only, the handler key stays).

Station	District	Challenges
CP Lab	Tech Quarter	ST-01 Receipt Total, SD-01 Peak Reading, SD-03 Robot Route, EX-01 Door and Key
Debugging Lab	Tech Quarter	ST-02 Broken Receipt, SD-04 Patch the Meter, EX-03 Ghost in the Counter
Logic Terminal	Tech Quarter	ST-03 Gate Cipher, SD-09 Fare Gate
Data Centre	Data Harbor	SD-02 Lonely Packet, EX-02 Chain of Command
Code Review Desk (was SQL Terminal)	Data Harbor	SD-05 Fence Count

Station	District	Challenges
Network Tower	Data Harbor	SD-08 Decode Chain
OSINT Terminal	Data Harbor	SD-10 Who Moved the Money
CTF Arena	Black Grid	SD-06 Password Profiler
Challenge Terminal	Black Grid	SD-07 Log Hunter
Sandbox Station	Black Grid	EX-04 Gatekeeper, EX-05 Portal Peek
Race Terminal	Central Exchange	RC-01, RC-02, RC-03
Raid Circle	Central Exchange	BR-01 to BR-10 (windows 1 and 2), BL-01 to BL-05 (window 3)
Home doors	Undercity	BG-01, BG-02, BG-03
Golden Hour Terminal and map events	NEXUS Plaza	GH-01, GH-02, GH-03

Each station also gives unpaid training. Training uses the same templates as the paid jobs of that station with new variants, and at most 6 training clears count per player.

Job boards

Board	Contracts
Tech Job Board	ST-01, ST-02, ST-03, SD-01, SD-03, SD-04, SD-09, EX-01, EX-03
Harbor Job Board	SD-02, SD-05, SD-08, SD-10, EX-02
Black Grid Job Board	SD-06, SD-07, EX-04, EX-05

Cards: Starter and Standard

Author copy: answers are included, so keep this document private. Every program below was compiled and run, and every answer comes from that run. Pay is Starter ₹10 and Standard ₹20 (see the money plan). The numbers, words and clues in each card are the example variant: the client regenerates them from the seed.

Starter

ST-01 Receipt Total (Code, C)

- Client: Veyra Transit Authority. Briefing: the fare gates logged a day of amounts, refunds are negative.
- Objective: read the program and type what it prints.
- Workbench: code panel, trace table.
- Answer: `105`.

```c
#include <stdio.h>
int main() {
    int amounts[6] = {40, -15, 25, -5, 30, 10};
    int total = 0;
    for (int i = 0; i < 6; i++) {
        if (amounts[i] > 0) {
            total = total + amounts[i];
        }
    }
    printf("%d\n", total);
    return 0;
}
```

ST-02 Broken Receipt (Debug, Python)

- Client: a junior clerk at the Transit Authority. Briefing: the total is too high because refunds are counted.
- Symptom shown: the program prints `85`, it should print `105`. One bug, on line 4.
- Fix options: `if a > 0:` (correct), `if a < 0:`, `if a == 0:`, `if a > 40:`.
- Run fixed code prints `105`.

```python
amounts = [40, -15, 25, -5, 30, 10]
total = 0
for a in amounts:
    if a != 0:
        total += a
print(total)
```

ST-03 Gate Cipher (CTF)

- Client: Black Grid gatekeeper. Briefing: the gate message is shifted and the header says by how much.
- Shown: `SHIFT: 3` and `DFFHVV JUDQWHG. IODJ: YHBUD{JDWH_RSHQ}`.
- Workbench: cipher bench with a shift wheel.
- Answer: shift each letter back by 3 to read `ACCESS GRANTED. FLAG: VEYRA{GATE_OPEN}`. Flag `VEYRA{GATE_OPEN}`.

Standard

SD-01 Peak Reading (Code, Java)

- Client: Veyra Power Grid. Briefing: six readings came in and the grid wants the busiest one.
- Answer: `Peak 45 at 3`.

```
public class Main {
    public static void main(String[] args) {
        int[] r = {12, 38, 7, 45, 30, 18};
        int best = r[0];
        int pos = 0;
        for (int i = 1; i < r.length; i++) {
            if (r[i] > best) {
                best = r[i];
                pos = i;
            }
        }
        System.out.println("Peak " + best + " at " + pos);
    }
}
```

SD-02 Lonely Packet (Code, Python)

- Client: Veyra Network Desk. Briefing: every packet should echo back, except the stray one.
- Answer: `5`. The first packet that appears once is 5 (6 also appears once but comes later).

```
packets = [4, 7, 4, 9, 7, 2, 9, 5, 2, 6]
for p in packets:
    if packets.count(p) == 1:
        print(p)
        break
```

SD-03 Robot Route (Code, C)

- Client: Veyra Courier Guild. Briefing: a delivery robot starts at 0 0 and follows a command string.
- Answer: `3 0`. Moves: R R U U L D R R D.

```
#include <stdio.h>
int main() {
    char cmd[] = "RRUULDRRD";
    int x = 0, y = 0;
    for (int i = 0; cmd[i] != '\0'; i++) {
        switch (cmd[i]) {
            case 'R': x = x + 1; break;
            case 'L': x = x - 1; break;
            case 'U': y = y + 1; break;
            case 'D': y = y - 1; break;
        }
    }
    printf("%d %d\n", x, y);
    return 0;
}
```

SD-04 Patch the Meter (Debug, C)

- Client: Veyra Utilities. Briefing: the meter prints running totals, and a reading of 0 means the meter was reset so the total restarts.
- Symptom shown: it prints `2 2 6 7`, it should print `3 5 0 4 5`. Two bugs.
- Bug 1: the loop starts at `i = 1`, so the first reading is skipped. Fix `int i = 0`.
- Bug 2: a zero reading does not reset the total. Fix: if `r[i] == 0` set `total = 0`, otherwise add it.

```
#include <stdio.h>
int main() {
    int r[5] = {3, 2, 0, 4, 1};
    int total = 0;
    for (int i = 1; i < 5; i++) {
        total = total + r[i];
        printf("%d ", total);
    }
    printf("\n");
    return 0;
}
```

SD-05 Fence Count (Debug, Java)

- Client: Veyra Parks Department. Briefing: a fence has 6 posts numbered 1 to 6, and the program should list them and count them.
- Symptom shown: it prints Post 1 to Post 5 and `Total 5`, it should end with `Total 6`. One bug, in the loop line.
- Fix options: `i <= posts` (correct), `i < posts - 1`, `i <= posts + 1`, `i != posts`.

```
public class Main {
    public static void main(String[] args) {
        int posts = 6;
        int count = 0;
        for (int i = 1; i < posts; i++) {
            System.out.println("Post " + i);
            count++;
        }
        System.out.println("Total " + count);
    }
}
```

SD-06 Password Profiler (CTF)

- Client: Black Grid Customs Office. Briefing: a dock worker left five clues about the gate word, and the manifest lists 30 candidates.
- Manifest: anchor, basket, candle, copper, dagger, ember, falcon, garnet, harbor, island, jacket, kettle, lantern, marble, needle, orchid, paddle, quartz, ribbon, saddle, timber, uplink, velvet, walnut, yonder, zephyr, bridge, cobalt, drift, flint.

- Clues in order (words left after each): starts with a consonant (25), is 6 letters long (22), has `a` as its second letter (11), contains `g` (2), ends with `t` (1).
- Workbench: elimination board. Answer: `VEYRA{garnet}`.

SD-07 Log Hunter (CTF)

- Client: Harbor Security Desk. Briefing: one account was breached last night and the login log has 50 to 60 lines.
- Objective: find the account that logged in successfully from a new IP within 2 minutes of 5 or more failures, and submit `VEYRA{account}`.
- Workbench: log viewer with search and filters.
- Build rule: one breached account, one decoy with 5 failures and no success (locked out), and one decoy with 1 failure then success from its usual IP. The example log has 53 lines and the rule above matches exactly one account.
- Answer: `VEYRA{rkhan}`.

```
2026-10-14 02:13:00 LOGIN FAIL user=rkhan ip=203.0.113.7
2026-10-14 02:13:08 LOGIN FAIL user=rkhan ip=203.0.113.7
2026-10-14 02:13:16 LOGIN FAIL user=rkhan ip=203.0.113.7
2026-10-14 02:13:24 LOGIN FAIL user=rkhan ip=203.0.113.7
2026-10-14 02:13:32 LOGIN FAIL user=rkhan ip=203.0.113.7
2026-10-14 02:13:40 LOGIN FAIL user=rkhan ip=203.0.113.7
2026-10-14 02:14:15 LOGIN OK user=rkhan ip=203.0.113.7
```

SD-08 Decode Chain (CTF)

- Client: Black Grid courier drop. Briefing: the message was wrapped three times.
- Shown: `}NUUJ_GHUAYN{YHAUD`.
- Workbench: cipher bench with reverse, Atbash and shift tools.
- Answer: reverse the text, apply Atbash (A becomes Z, B becomes Y), then shift back by 1. Flag `VEYRA{LAYERS_PEEL}`.

SD-09 Fare Gate (Puzzle)

- Client: the Transit Authority turnstile panel. Briefing: the gate is jammed. Pressing a light toggles it and its neighbours up, down, left and right. Turn all 9 lights off.
- Start board (X is on): `XX.`, `X.X`, `XXX`. Generated from 3 presses, so it is always solvable in 3 or fewer.
- Answer: press row 1 column 2, row 2 column 3 and row 3 column 1.

SD-10 Who Moved the Money (Puzzle)

- Client: Veyra Fraud Desk. Briefing: four suspects, four statements, exactly one lies.
- Statements: Aman says "Bela did it." Bela says "Chitra did it." Chitra says "Bela is lying." Dev says "I did not do it."
- Answer: Bela. Any other thief leaves two or three liars.

Cards: Expert

Expert pays ₹80 and has one slot per template. It asks for more steps, not a harder topic.

EX-01 Door and Key (Puzzle)

- Client: Veyra Vault Services. Briefing: the exit is behind a locked door `D` and the key `K` is somewhere on the map.
- Play: move a token with the arrow keys. The door opens once the key is picked up. The page accepts the run when the token reaches `E` in the fewest possible steps.
- Answer: 10 steps. One shortest route is R R D D R R D D R D. Without the key the exit is unreachable.

```
S..#..
.#.#.#
.#K...
.###D#
.#....
####.E
```

EX-02 Chain of Command (Puzzle)

- Client: Veyra Staffing Office. Briefing: HR wants the two employees who are furthest apart in the company.
- Play: the app draws the tree below. The player taps the two ends of the longest chain. The page counts the hops.
- Tree: Asha is the CEO. Bimal and Chitra report to Asha. Dev and Esha report to Bimal. Farid reports to Chitra. Gita and Hari report to Dev. Isha reports to Farid.
- Answer: Gita and Isha, 6 hops (Gita, Dev, Bimal, Asha, Chitra, Farid, Isha).

EX-03 Ghost in the Counter (Debug, C)

- Client: Veyra Courier Guild, second shift. Briefing: the program counts readings above 50 and adds them up, but the answer is wrong.
- Symptom shown: it prints `5 271`, it should print `3 221`. Three bugs.
- Bug 1: `int count = 1;` should start at 0.
- Bug 2: the loop uses `i < 5` and skips the last reading. It should be `i < 6`.
- Bug 3: `>= 50` counts the reading of exactly 50. It should be `> 50`.

```
#include <stdio.h>
int main() {
    int v[6] = {72, 15, 88, 50, 61, 33};
    int count = 1;
    int sum = 0;
    for (int i = 0; i < 5; i++) {
        if (v[i] >= 50) {
            count++;
            sum = sum + v[i];
        }
    }
    printf("%d %d\n", count, sum);
    return 0;
}
```

EX-04 Gatekeeper (CTF)

- Client: Black Grid door terminal. Briefing: the door turns your key into a code, and it only opens on code 20. Find the key.
- Machine: take the key, multiply by 3, add 8, divide by 2, subtract 5. The result is the code.
- Workbench: number machine with 5 test runs and the calculator. The player works backwards from 20.
- Answer: key `14` (14 x 3 = 42, plus 8 is 50, halved is 25, minus 5 is 20). The door then shows `VEYRA{GATEKEEPER}`.

EX-05 Portal Peek (CTF)

- Client: a mock "Veyra Bank" staff portal, clearly labelled as a training sandbox. Briefing: log in as admin. The staff notice board says the password is hidden in plain sight.

- Notices, in order: "Vault hours are 9 to 5 on weekdays." "All tellers wear a blue badge." "Updates to the app ship on Fridays." "Lost cards: call the front desk." "Tea is served in the lobby at 4."
- Answer: the first letters spell the password `VAULT`. Log in as `admin` with `VAULT` to see `VEYRA{FIRST_LETTERS}`.
- Build note: everything is drawn inside the challenge screen, with no real login and no network calls.

Cards: Burglary, Races and Golden Hour

Burglary lock-picks (Undercity Home doors)

The thief faces the lock on another player's Home. The amount stolen is the smallest of the share of the Home money, the level maximum, and the Home money minus ₹5. A failed or timed-out attempt costs the thief ₹3 (to the Treasury) and 3 minutes in jail.

Level	Lock	Share	Maximum	Puzzle time	The thief needs
1 easy	BG-01 Code Oracle	5%	₹10	2 to 3 minutes	Nothing
2 hard	BG-02 Maze Lock	10%	₹25	4 to 5 minutes	5 completed challenges
3 difficult	BG-03 Circuit Lock	20%	₹50	6 to 8 minutes	12 completed challenges

BG-01 Code Oracle (Puzzle)

- Briefing: a keypad on the target's door. Guess the 3-digit code. Each guess shows how many digits are right and in place, and how many are right but in the wrong place. Digits are all different. 9 guesses, then the alarm triggers.
- Example: secret 284, guess 482 shows 1 in place and 2 misplaced.

BG-02 Maze Lock (Puzzle)

- Briefing: a corridor behind the door. Steer from `S` to `E` with U, D, L, R. The door `D` stays shut until you step on the key `K`.
- Answer: `DDRRUDRRD` (9 moves). The page simulates the string, so any valid route counts.

```
S.#..
.#K#.
...D.
##.#E
```

BG-03 Circuit Lock (Puzzle)

- Briefing: a fuse box with 6 switches A to F and 5 gates. Make the output lamp light.
- Gates: G1 = A AND B. G2 = C OR D. G3 = G1 XOR G2. G4 = E OR F. Output = G3 AND G4.
- Answer: any of 30 settings works. Example: only C and E on.

Races (Race Terminal, pool ₹100 each: ₹50, ₹30, ₹20)

Every entered player gets a different variant of the same difficulty. The window is 10 minutes and the server ranks the first correct results.

RC-01 Sprint: Fare Sum (Code, Python)

- Briefing: the turnstile logged some fares. Type what the program prints. Answer for this variant: `48`.

```
fares = [12, 5, 8, 20, 3]
total = 0
for f in fares:
    total += f
print(total)
```

RC-02 Sprint: Bug Hunt (Debug, C, Python or Java)

- Briefing: `n` is 5 and the program should add the numbers from 1 to n, which is 15. It prints 10. Each player gets the same bug in C, Python or Java: the loop starts at 0 and stops one early.
- Python fix: `range(1, n + 1)`. C and Java fix: `int i = 1; i <= n`. All three versions were run.

```
n = 5
total = 0
for i in range(n):
    total += i
print(total)
```

RC-03 Sprint: Decode (CTF)

- Briefing: the message was wrapped twice.
- Shown: `}F_YSNWUX{FWDJA`. Peel: reverse the text, then shift back by 5. Flag `VEYRA{SPRINT_A}`. Change the flag text and shift per player.

Golden Hour (NEXUS Plaza, pool ₹100 each)

GH-01 Key Card (CTF, about 1 minute)

- Play: 5 key cards appear on random open tiles, with a NEXUS clue each minute. A player taps a card and solves its decode. The first 5 correct players get ₹20 each, one card per player.
- Shown: `}PL_WIZX_BVP{ZIBVE`. Peel: reverse, then Atbash. Flag `VEYRA{KEY_CARD_OK}`.

GH-02 Courier Interrogation (Puzzle)

- Play: a courier walks the map for 10 minutes. A player next to the courier gets the lock, then guesses a number from 1 to 100 with 8 guesses, the courier answering higher or lower. A failed player releases the lock to the next one. The first correct player gets ₹100.
- Note: a binary search needs at most 7 guesses, so 8 is fair.

GH-03 Server Sprint (Code, Python, 2 to 3 minutes)

- Play: all players at the Golden Hour Terminal get a short challenge in a 5-minute window. The top 5 get ₹40, ₹25, ₹15, ₹10 and ₹10.
- Briefing: servers report UP or DOWN in order. Type what the program prints. Answer: `3`.

```
status = ["UP", "UP", "DOWN", "UP", "UP", "UP", "DOWN", "UP"]
best = 0
run = 0
for s in status:
    if s == "UP":
        run += 1
        if run > best:
            best = run
    else:
        run = 0
print(best)
```

Cards: Raid questions

Each raider gets 5 questions: 2 from tier A, 2 from tier B and 1 from tier C. A raider must solve at least 2 of 5 to pass the gate, and the raid succeeds when at least 5 of the 10 pass. The value of one solved question is the raid prize divided by all solved questions of passing raiders, capped at ₹160. Tier A is kept easy so that 50 to 60% of the crew pass the gate. If playtests show fewer, make tier A easier and do not change the pay.

Tier A

BR-01 Tunnel Network (Puzzle)

- Briefing: digits in the grid are teleport pads. Standing on a pad, you may jump to the other pad with the same digit as one step. Reach `E` in the fewest steps.
- Play: move a token. Answer: 7 steps (S, right, right onto pad 1, jump to the other pad 1, then right, right, right, right to E). Pad 2 is a decoy.

```
S.1#2..
.######
..1.2.E
```

BR-02 Double Login (Code, Java)

- Briefing: the bank flags a user who logs in two or more times. Type what the program prints. Answer: `DOUBLE 3`.

```
public class Main {
    public static void main(String[] args) {
        String[] log = {"ana", "raj", "dev", "raj", "ana", "raj"};
        int count = 0;
        for (int i = 0; i < log.length; i++) {
            if (log[i].equals("raj")) {
                count++;
            }
        }
        if (count >= 2) {
            System.out.println("DOUBLE " + count);
        } else {
            System.out.println("OK");
        }
    }
}
```

BR-03 Cipher Relay (CTF)

- Briefing: each letter was shifted forward by its position in the message: the 1st letter by 1, the 2nd by 2, and so on. Only letters are counted. Braces and underscores stay.
- Shown: `WGBVF{XLTJI_CGA}`. Peel: shift each letter back by its position. Flag `VEYRA{RELAY_RUN}`.

BR-04 Ghost Log (CTF)

- Briefing: one session was never logged out and was active at 3 AM. The log has 50 to 60 lines. Find the session ID.
- Build rule: every session has a LOGIN and a LOGOUT except one, which has only a LOGIN and two ACTION lines at about 03:12. The rule was checked on a 43-line base log with exactly one open session; pad with 8 to 15 ordinary sessions to reach 50 to 60 lines.
- Workbench: log viewer. Answer: `VEYRA{S-4471}`.

```
03:12:00 LOGIN  session=S-4471 user=omar
03:14:00 ACTION session=S-4471 export_all_records
03:15:00 ACTION session=S-4471 clear_audit_trail
```

Tier B

BR-05 Org Chart (Puzzle)

- Briefing: the app draws the company tree from EX-02 (Asha at the top, Bimal and Chitra under her, and so on). Find the nearest boss that two employees share.
- Play: tap the shared boss. Answer for Gita and Esha: Bimal. Other pairs: Hari and Isha share Asha.

BR-06 Combination Clues (CTF)

- Briefing: a 4-digit code with all different digits. Each guess shows how many digits are right and in place, and how many are right but in the wrong place.
- Clues: 7314 gives 1 in place and 2 misplaced. 9081 gives 0 and 2. 7509 gives 1 and 1. 7308 gives 0 and 1. 8234 gives 0 and 1.
- Workbench: elimination board with digit slots. Answer: `4719`, flag `VEYRA{4719}`. This is the only code that fits.

BR-07 Self-Checking Vault (CTF)

- Briefing: a 4-digit key with all different digits. The first digit is double the last. The second digit is the smallest of the four. The third digit is bigger than the first. The four digits add up to 20.
- Answer: `6293`, flag `VEYRA{6293}`. This is the only key that fits.

BR-08 Broken Tally (Debug, C)

- Briefing: the program counts cells above the limit 6 in each row of a grid and prints one count per row. Two bugs.
- Symptom shown: it prints `2 4 6`, it should print `2 1 2`.
- Bug 1: `count` is declared before the row loop, so it never resets. Move `int count = 0;` inside the row loop.
- Bug 2: `>=` counts cells equal to the limit. It should be `>`.

```
#include <stdio.h>
int main() {
    int g[3][4] = {{5, 9, 2, 7}, {1, 8, 6, 3}, {9, 9, 4, 1}};
    int limit = 6;
    int count = 0;
    for (int r = 0; r < 3; r++) {
        for (int c = 0; c < 4; c++) {
            if (g[r][c] >= limit) {
                count++;
            }
        }
        printf("%d ", count);
    }
    printf("\n");
    return 0;
}
```

Tier C

BR-09 Deploy Order (Puzzle)

- Briefing: six tasks A to F. Rules: B before A, C before A, E before C, D before F, B before D, E before B. Drag them into the smallest valid order, taking the earliest letter whenever several tasks are ready.
- Answer: E, B, C, A, D, F.

BR-10 Constraint Vault (Puzzle)

- Briefing: give each of A to F a different number from 1 to 6 so all 8 conditions hold. A > B. C is odd. D + B = A. E > C. F is even. The difference between A and F is 1. C > A. B is odd.
- Answer: A=3, B=1, C=5, D=2, E=6, F=4. It is the only solution.

Cards: Black Ledger

The Black Ledger is the final raid (window 3, minutes 150 to 175). Every raider gets all 5 questions, the gate is still 2 of 5, and the prize is the Black Ledger seed plus anything rolled forward. The dark ledger-book theme makes it feel like a higher tier.

BL-01 Ledger Running Total (Code, Java)

- Briefing: the ledger applies deposits and withdrawals in order, but ignores any withdrawal that would take the balance below zero. Type what the program prints. Answer: `75`.

```
public class Main {
    public static void main(String[] args) {
        int[] moves = {50, -20, 30, -100, 40, -25};
        int balance = 0;
        for (int i = 0; i < moves.length; i++) {
            if (balance + moves[i] >= 0) {
                balance = balance + moves[i];
            }
        }
        System.out.println(balance);
    }
}
```

BL-02 Triple Cipher (CTF)

- Briefing: the courier wrapped the message three times.
- Shown: `}MUIL_SLHOFD{WFYSB`. Peel: reverse the text, apply Atbash, then shift back by 3. Flag `VEYRA{TRIPLE_LOCK}`.

BL-03 Shadow Ring (Puzzle)

- Briefing: the app draws six accounts with payment arrows. A ring is a loop of payments that returns to where it started. Tap the accounts that form the ring.
- Payments: A to B, B to C, C to D, D to B, C to E, E to F.
- Answer: the ring B, C, D. It is the only loop.

BL-04 Warden Script (CTF)

- Briefing: the warden takes your number and repeats two steps three times: double it, then subtract 1. The vault opens only if the final number is 17. Find the number to start with.
- Workbench: number machine with 5 test runs and the calculator.
- Answer: start `3` (3 becomes 5, then 9, then 17). Flag `VEYRA{WARDEN_DOWN}`.

BL-05 Broken Router (Debug, Python)

- Briefing: the program picks the cheapest of 5 routes and prints which one and its cost. Three bugs.
- Symptom shown: it prints `Route 1 costs 52`, it should print `Route 4 costs 29`.
- Bug 1: `best = 0` should be `best = costs[0]`.
- Bug 2: `>` should be `<`.
- Bug 3: the winning index is never saved. Add `best_index = i` under the update.

```
costs = [48, 35, 52, 29, 41]
best = 0
best_index = 0
for i in range(1, len(costs)):
    if costs[i] > best:
        best = costs[i]
print("Route", best_index + 1, "costs", best)
```

Money plan for 30 to 40 players

The world still holds exactly ₹4,000 (400,000 paise), sized for 40 players. Two rules differ from the build specification: the Bank pays no interest and instead charges a holding fee that feeds raid prizes, and a player who reaches the ₹800 cap stays in the game. All values below are proposals and are editable in the spreadsheet, which recalculates the totals.

Accounts at the start

Account	Calculation	₹
Starter jobs	40 claims x ₹10	400
Standard jobs	5 waves x 8 slots x ₹20	800
Expert jobs	5 jobs x ₹80	400
Races	3 races x ₹100	300
Golden Hour	3 events x ₹100	300
Bank raid seed, window 1	Minutes 45 to 90	400
Bank raid seed, window 2	Minutes 90 to 135	400

Account	Calculation	₹
Black Ledger seed, window 3	Minutes 150 to 175	700
Treasury	Failed burglary fees and wealth bounties	300
Bank fee vault	Starts empty and fills with holding fees	0
Total		4,000

Compared with the specification: Starter is smaller (40 claims instead of 90), Standard pays ₹20 instead of ₹10 with 40 slots instead of 45, and the ₹300 interest pool is gone. The net change is zero.

Standard waves (8 slots each, 4 per template)

Wave	Minute	Template (station)	Template (station)
1	0	SD-01 Peak Reading (CP Lab)	SD-02 Lonely Packet (Data Centre)
2	20	SD-04 Patch the Meter (Debugging Lab)	SD-06 Password Profiler (CTF Arena)
3	50	SD-03 Robot Route (CP Lab)	SD-08 Decode Chain (Network Tower)
4	80	SD-05 Fence Count (Code Review Desk)	SD-09 Fare Gate (Logic Terminal)
5	120	SD-07 Log Hunter (Challenge Terminal)	SD-10 Who Moved the Money (OSINT Terminal)

Expert jobs keep the specification schedule: EX-01 at minute 30, EX-04 at 60, EX-02 at 90, EX-05 at 120 and EX-03 at 140, one slot each at ₹80.

Bank holding fee

- The Bank opens at minute 10. Every minute that money sits in the Bank, the Bank charges 0.1% of the balance, rounded to the nearest paisa. Money in transit is not charged.
- Fees go to the Bank fee vault. The next raid prize is the window seed plus the vault plus 10% of all Bank and in-transit money. Fees only move money, so the total stays ₹4,000.

- The maximum Bank balance per player stays ₹200.
- If a window ends with no raid, the vault carries over to the next one. Vault money left at the freeze stays with the organizers.

Balance held	30 minutes	60 minutes	120 minutes
₹50	₹1.48	₹2.91	₹5.66
₹100	₹2.96	₹5.83	₹11.31
₹200	₹5.91	₹11.65	₹22.63

The tension is meant: ₹200 at the Home can lose up to ₹40 to one difficult burglary, and keeping it safe in the Bank for an hour costs about ₹12.

Cap, exits and the most one player can earn

- At ₹800 (Home plus Bank) a player keeps playing. A payment that would go above the cap is trimmed, and the trimmed part stays where it came from: the pool, the victim's Home, or the raid prize. Stolen money is trimmed the same way.
- Checkout opens at minute 150 (proposed). A player who is disconnected for 10 minutes is still checked out automatically.
- The most one player can earn from challenge pools is ₹820: Starter ₹10, Standard ₹100 (one slot in each of 5 waves), Expert ₹400, races ₹150 and Golden Hour ₹160. The cap trims the extreme case.
- With 30 players instead of 40, up to ₹100 of the Starter pool stays unclaimed and goes back to the organizers. Nothing else changes.

Total rule: Home + Bank + in transit + all pools + seeds + Treasury + fee vault + Payout equals 400,000 paise at all times, and the server checks it every 5 seconds.

Showing it on the map

The map is a static pixel-art scene, so do not repaint it. Add a status layer on top, positioned from the coordinates in `art-world.json`. It works for every hub and costs little.

Terminal markers

- One colour per type so players learn it in minutes: cyan for Code, red for Debug, green for CTF, violet for Puzzle. The art is mostly warm amber, so these stand out.
- A small glyph above each station (`</>` , bug, flag, puzzle piece) with 1 to 3 difficulty pips for Starter, Standard and Expert.
- States: pulsing means a slot is open, a solid ring means someone is using it, grey with a timer means cooldown, a check mark means cleared.
- Pressing **M** can filter the map to unfinished contracts only.

Challenge screens

Skin the puzzles so they look like objects in the city.

Challenge	Skin
SD-09 Fare Gate	A turnstile panel with a 3x3 light grid
BG-01 Code Oracle	A keypad on a door
BG-02 Maze Lock	A corridor with a key and a door
BG-03 Circuit Lock	A fuse box with wires and gates
EX-05 Portal Peek	A mock staff notice board and login window
GH-02 Courier Interrogation	A dialogue box with a courier sprite and "higher" or "lower" replies

Events and money on the map

Moment	Visual
Away from Home	A red "Home exposed" dot on the HUD whenever the player is at a station, since that is when a thief can strike
Bank holding fee	A small fee ticker next to the Bank balance on the HUD, so the cost of waiting is always visible

Moment	Visual
Burglary	A lock-ring icon on the Home door while someone is picking. The owner gets a message with the amount and the thief's name afterwards
Raid Circle	10 slots around the ring that light up as operators stand in. Amber while assembling, red during the raid, with the 60-second hold as a progress arc
Bank during a raid	Shutters or a red alarm tint on the facade, and a "Closed for raid" sign
Wealth alert	A gold flash on the NEXUS ticker when one Home passes ₹300
Golden Hour	A gold ring at the NEXUS terminal, a faint gold vignette, and a countdown in the top bar. GH-01 key cards glow on their tiles
Races	A start-gate glow at the Race Terminal, a countdown strip across the top, and live finishers on the Leaderboard
Black Ledger	The dark ledger-book theme on the challenge screens

Announcements

Use the existing NEXUS ticker for everything: "Wave 3 opens", "Race starting at Central Exchange", "Raid crew forming: 6 of 10", "Golden Hour opens in 2 minutes". It is the cheapest way to make the world feel alive.

Open decisions

Each row has my proposal. Anything you do not change is built as proposed.

#	Topic	Proposal
1	Early exit	Checkout opens only at minute 150, because leaving early is the same hole as retiring at the cap

#	Topic	Proposal
2	Bank fee	0.1% of the balance per minute, maximum ₹200 in the Bank, fees feed the next raid prize. Tune the rate in the spreadsheet
3	Raid crew size	Keep 10. Windows 1 and 2 need 20 different players, which is two thirds of a 30-player room. Drop the crew to 8 if fewer than 36 players sign up
4	SQL Terminal	Relabel it "Code Review Desk" (label only)
5	Homes	Use wards A to C (45 addresses) for up to 40 players, so empty Homes do not dilute burglary
6	Prize	The average final balance is about ₹100 at 40 players and about ₹133 at 30. Decide the minimum payout and how it is paid
7	Cap	₹800 is 20% of the world. Lower it to ₹600 if you want the prizes spread more evenly
8	Checking	Answers are checked in the browser, so add a minimum solve time per challenge and review claim times after the event
9	Outside help	AI tools cannot be blocked in a browser game. Announce a rule, and draw code as images so it is harder to paste